

Notes: Dell (Econometrica, 2010)

The Persistent Effects of Peru's Mining Mita

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
2.1	Contribution relative to the historical institutions literature	2
2.2	Contribution relative to mechanisms of persistence	2
2.3	Contribution relative to regression discontinuity applications	3
3	Historical background and institutional setting	3
3.1	The mining mita	3
3.2	Geographic boundary	3
4	Data and measurement	4
4.1	Modern living standards	4
4.2	Historical validity checks	4
4.3	Mechanism variables	4
5	Empirical specifications	4
5.1	Baseline geographic RD	4
5.2	Alternative RD polynomials	5
5.3	Mechanism regressions	5
6	Identification and validity	5
6.1	Identifying assumption	5
6.2	Balance tests	5
6.3	Limitations	6
7	Tables and figures	7
7.1	Figure 1: Mita boundary	7
7.2	Table I: Summary statistics and boundary balance	7
7.3	Table II and Table III: Main effects and robustness	8
7.4	Figure 2: Spatial RD plots of outcomes	9
7.5	Table V and Table VI: Pre-treatment data and land tenure	10
7.6	Table VII and Table VIII: Education and roads	11
7.7	Table IX and Figure 3: Consumption channels and market participation	12
8	Short summary	13
9	Conclusion	13
9.1	Core conclusion	13
9.2	Economic intuition	14
9.3	Takeaway	14

1 Big picture

- **Question.** Do coercive colonial institutions have persistent effects on modern economic outcomes, and through what channels do those effects survive for centuries?
- **Historical institution.** The mining mita was a forced labor draft imposed by the Spanish colonial state in Peru and Bolivia from 1573 to 1812. Indigenous communities inside the mita region were required to send roughly one-seventh of adult men to the Potosi silver and Huancavelica mercury mines.
- **Empirical design.** The paper uses a geographic regression discontinuity design around the mita boundary. Communities just inside and outside the boundary are compared, with flexible controls for latitude, longitude, distance to Potosi, distance to the boundary, and boundary-segment fixed effects.
- **Main modern outcomes.** The mita reduces modern equivalent household consumption by about 25 percent and raises childhood stunting by about 6 percentage points in subjected districts.
- **Mechanism.** The mita altered land tenure and public-good provision. It limited the formation of haciendas inside the mita region, preserved communal land tenure during the colonial period, and left these areas with weaker post-colonial property-rights systems and less public infrastructure.
- **Historical channels.** Mita districts historically had fewer haciendas, lower educational attainment, and weaker integration into transportation networks.
- **Contemporary channels.** Mita districts have lower road density, more subsistence agriculture, less market participation, and greater reliance on agricultural labor.
- **Core claim.** Extractive historical institutions can generate persistent poverty not only by reducing contemporaneous output, but also by reshaping local property-rights regimes and the incentives for public-goods investment.
- **Takeaway.** The paper turns a broad historical-institutions hypothesis into a micro-geographic test with explicit evidence on mechanisms of persistence.

2 Incremental contribution of the paper

2.1 Contribution relative to the historical institutions literature

- Earlier work, especially cross-country studies, documented that colonial institutions are correlated with modern income, property rights, and development.
- Dell's incremental contribution is to use within-country geographic variation in a single historical institution, avoiding many confounds that affect cross-country comparisons.
- The mita boundary creates a discontinuous treatment rule in an otherwise geographically continuous region, allowing a regression discontinuity design.
- This makes the paper one of the canonical micro-level studies of colonial institutional persistence.

Interpretation: The paper moves the literature from macro historical correlations toward quasi-experimental identification of one institution's long-run effects.

2.2 Contribution relative to mechanisms of persistence

- Many studies show that history matters but do not distinguish between possible mechanisms.
- Dell traces channels through historical and modern data: 1572 tribute data, 1689 hacienda data, 1845 and 1940 hacienda measures, education, road networks, and agricultural production.

- The evidence points to land tenure and public goods rather than only ethnicity, geography, or pre-treatment differences.
- The paper therefore explains not only *whether* the mita mattered, but also *how* its effects persisted.

Interpretation: The mechanism evidence is central. The paper does not stop at a reduced-form treatment effect; it uses multiple historical datasets to reconstruct the institutional path from coercive labor to modern poverty.

2.3 Contribution relative to regression discontinuity applications

- The paper applies a multidimensional geographic RD to a historical boundary.
- It addresses the problem that geography changes in two dimensions, not one, by exploring alternative RD polynomials in latitude/longitude, distance to Potosi, and distance to the boundary.
- It also tests for discontinuities in pre-treatment variables such as elevation, slope, indigenous share, tribute, and colonial administrative allocations.
- This makes the paper a template for later historical RD studies using political or institutional boundaries.

Interpretation: The paper’s empirical contribution is methodological as well as substantive: it shows how to make historical boundaries credible for causal inference.

3 Historical background and institutional setting

3.1 The mining mita

- The Spanish colonial administration introduced the mita in 1573.
- The institution required indigenous communities in a defined region to send laborers to mines, especially Potosi and Huancavelica.
- Drafted workers faced harsh conditions, long travel, low pay, and high mortality.
- The mita was abolished in 1812, but its effects are hypothesized to persist through local institutional structures.

Interpretation: The mita was not a small labor tax. It was a large coercive extraction system that reallocated labor and altered settlement, production, and land-tenure incentives.

3.2 Geographic boundary

- The assignment rule was spatial: communities inside the boundary were treated, and communities outside were exempt.
- The paper focuses on southern Peru, where the boundary traces the Andean range and where observable geography changes relatively smoothly.
- The main estimation sample includes observations within 100, 75, 50, and sometimes 25 kilometers of the boundary.

Interpretation: The geographic discontinuity is the basis for identification. The central question is whether units just inside and outside the boundary were otherwise comparable.

4 Data and measurement

4.1 Modern living standards

- **Household consumption:** measured using the 2001 Peruvian household survey. The main variable is log equivalent household consumption.
- **Child stunting:** measured using a 2005 national height census for children, typically ages 6–9 in the baseline table and younger children in some robustness exercises.
- Consumption is a household-level measure, while stunting provides an independent health-based measure of living standards.

Interpretation: Using both consumption and child height reduces the concern that the result is driven by one survey or one measurement system.

4.2 Historical validity checks

- The paper collects 1572 tribute data from before the mita’s enactment.
- It also collects variables such as elevation, slope, population measures, ethnic composition, and colonial administrative variables.
- These variables are used to test whether the boundary already separated systematically different districts before treatment.

Interpretation: If pre-treatment characteristics changed discontinuously at the boundary, the RD interpretation would be weakened. Table I and Table V are therefore key validation exercises.

4.3 Mechanism variables

- **Land tenure:** haciendas per district in 1689, haciendas per thousand residents in 1689, shares of rural population living in haciendas in 1845 and 1940, and land Gini in 1994.
- **Education:** literacy in 1876, mean schooling in 1940, and mean schooling in 2001.
- **Roads:** 2006 densities of local road networks, regional road networks, and paved/gravel regional roads.
- **Consumption channels:** agricultural labor force share, agricultural households selling output in markets, and employment outside agriculture.

Interpretation: The mechanism variables let the paper distinguish a pure historical income effect from specific channels of persistence.

5 Empirical specifications

5.1 Baseline geographic RD

A generic version of the baseline specification is

$$y_{id} = \alpha + \beta Mita_d + f(Geo_d) + X_d' \gamma + \lambda_s + \varepsilon_{id}, \quad (1)$$

where y_{id} is an individual, household, or district outcome; $Mita_d$ equals one for districts inside the mita region; $f(Geo_d)$ is a flexible geographic polynomial; X_d contains geographic controls; and λ_s are boundary-segment fixed effects.

Interpretation: The coefficient β is the estimated discontinuity at the mita boundary. It compares locations very near each other but on opposite sides of the historical treatment boundary.

5.2 Alternative RD polynomials

The paper considers several alternatives:

$$y_d = \alpha + \beta Mita_d + f_1(Lat_d, Lon_d) + X_d' \gamma + \lambda_s + \varepsilon_d, \quad (2)$$

$$y_d = \alpha + \beta Mita_d + f_2(DistPotosi_d) + X_d' \gamma + \lambda_s + \varepsilon_d, \quad (3)$$

$$y_d = \alpha + \beta Mita_d + f_3(DistBoundary_d) + X_d' \gamma + \lambda_s + \varepsilon_d. \quad (4)$$

- The paper varies whether f is linear, quadratic, quartic, or cubic.
- It also considers specifications that interact polynomials with the mita treatment indicator.
- The preferred baseline often uses cubic polynomials and boundary-segment fixed effects.

Interpretation: The robustness of the coefficient across polynomial choices is important because the boundary is multidimensional, not a simple one-dimensional cutoff.

5.3 Mechanism regressions

Mechanism regressions use the same RD structure but change the dependent variable:

$$Mechanism_d = \alpha + \beta Mita_d + f(Geo_d) + X_d' \gamma + \lambda_s + \varepsilon_d. \quad (5)$$

Interpretation: A significant mita coefficient on land tenure, roads, education, or market participation suggests that the mita affected these intermediate channels.

6 Identification and validity

6.1 Identifying assumption

- The RD identifying assumption is that potential outcomes vary smoothly through the mita boundary in the absence of treatment.
- The mita boundary must not coincide with other discontinuities that independently affect modern development.
- To make this plausible, the paper focuses on a part of the boundary where geography and ethnicity are relatively smooth.

Interpretation: The key comparison is not mita Peru versus all non-mita Peru. It is districts near the boundary, weighted by their proximity and geographic controls.

6.2 Balance tests

- Table I shows that elevation, slope, indigenous share, tribute rates, and colonial administrative variables are generally similar inside and outside the boundary.
- Table V uses 1572 data and finds no systematic pre-treatment differences in tribute or population composition.
- These balance tests reduce concerns that the boundary captured pre-existing differences rather than the mita.

Interpretation: The pre-treatment evidence is the main defense against the objection that poor regions were simply selected into the mita.

6.3 Limitations

- Historical data are incomplete and sometimes measured at coarse geographic levels.
- Treatment status may be correlated with unobserved geography despite extensive controls.
- Some mechanisms, such as political power and social conflict, are difficult to measure directly.
- The RD identifies effects near the boundary, not necessarily the average effect over all mita districts.

Interpretation: The paper is strongest for local causal inference near the mita boundary and for documenting plausible channels of persistence.

7 Tables and figures

The visuals below are directly cropped from the original paper. Nearby figures and tables are placed on the same row where possible, and single figures/tables use width $0.6nlinewidth$.

7.1 Figure 1: Mita boundary

Read:

- The black line is the mita boundary, and the light gray line is the study boundary.
- Districts inside the contiguous area formed by the mita boundary were required to contribute labor to the mita.
- Elevation is shown in the background, emphasizing the importance of geographic controls.

Economic message: The map makes the treatment assignment visible. The identification strategy depends on comparing locations just inside and outside this boundary.

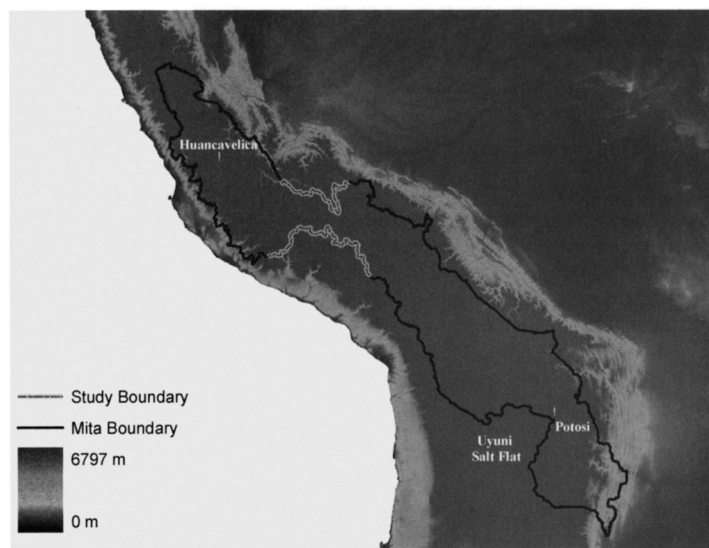


FIGURE 1.—The *mita* boundary is in black and the study boundary in light gray. Districts falling inside the contiguous area formed by the *mita* boundary contributed to the *mita*. Elevation is
Figure 1: Mita boundary and study boundary

7.2 Table I: Summary statistics and boundary balance

Read:

- Table I compares districts inside and outside the mita boundary at several distance bands: 100, 75, 50, and 25 kilometers.
- Elevation, slope, indigenous share, and tribute variables are generally similar near the boundary.
- The balance evidence supports the claim that the boundary does not separate sharply different pre-treatment regions.

Economic message: The table is a validation exercise. If geography and colonial baseline characteristics were discontinuous, the RD design would be less credible.

	Sample Falls Within											
	<100 km of <i>Mita</i> Boundary			<75 km of <i>Mita</i> Boundary			<50 km of <i>Mita</i> Boundary			<25 km of <i>Mita</i> Boundary		
	Inside	Outside	s.e.	Inside	Outside	s.e.	Inside	Outside	s.e.	Inside	Outside	s.e.
GIS Measures												
Elevation	4042	4018	[188.77] (85.54)	4085	4103	[166.92] (82.75)	4117	4096	[169.45] (89.61)	4135	4060	[146.16] (115.15)
Slope	5.54	7.21	[0.88]* (0.49)**	5.75	7.02	[0.86] (0.52)**	5.87	6.95	[0.95] (0.58)*	5.77	7.21	[0.90] (0.79)*
Observations	177	95		144	86		104	73		48	52	
% Indigenous	63.59	58.84	[11.19] (9.76)	71.00	64.55	[8.04] (8.14)	71.01	64.54	[8.42] (8.43)	74.47	63.35	[10.87] (10.52)
Observations	1112	366		831	330		683	330		329	251	
Log 1572 tribute rate	1.57	1.60	[0.04] (0.03)	1.57	1.60	[0.04] (0.03)	1.58	1.61	[0.05] (0.04)	1.65	1.61	[0.02]* (0.03)

Table I, first page

km of <i>Mita</i> Boundary	Sample Falls Within											
	<75 km of <i>Mita</i> Boundary			<50 km of <i>Mita</i> Boundary			<25 km of <i>Mita</i> Boundary			<25 km of <i>Mita</i> Boundary		
	Inside	Outside	s.e.	Inside	Outside	s.e.	Inside	Outside	s.e.	Inside	Outside	s.e.
63.82	[1.39]*** (1.36)***	59.98	63.69	[1.56]** (1.53)**	62.01	63.07	[1.12] (1.34)	61.01	63.17	[1.58] (2.21)		
19.10	[0.90]** (0.94)**	21.90	19.45	[1.02]** (1.02)**	20.59	19.93	[0.76] (0.92)	21.45	19.98	[1.01] (1.33)		
12.58	[0.53] (0.48)*	13.31	12.46	[0.65] (0.60)	12.81	12.48	[0.43] (0.55)	13.06	12.37	[0.56] (0.79)		
4.40	[0.78] (0.85)	4.55	4.29	[0.26] (0.29)	4.42	4.47	[0.34] (0.33)	4.48	4.42	[0.29] (0.39)		
41		47	37		35	30		18	24			

grid cells for the geospatial measures, the household for % indigenous, and the district for the 1572 tribute data. Conley standard errors and non-*mita* observations are in brackets. Robust standard errors for the difference in means are in parentheses. For % indigenous, or clustering at the district level. The geospatial measures are calculated using elevation data at 30 arc second (1 km) resolution (SRTM 1 is 1000 meters and for slope is degrees. A household is indigenous if its members primarily speak an indigenous language in the home taken from Miranda (1583). In the first three columns, the sample includes only observations located less than 100 km from the *mita* > 75, 50, and finally 25 km in the succeeding columns. Coefficients that are significantly different from zero are denoted by the following

Table I, continued

7.3 Table II and Table III: Main effects and robustness

Read:

- Table II estimates the *mita*'s effect on modern living standards.
- The consumption coefficients are negative, with preferred specifications implying roughly a 25 percent reduction in equivalent household consumption.
- The stunting coefficients are positive, often around 6 percentage points or more, showing worse child health in *mita* districts.
- Table III shows robustness to many RD polynomial choices and estimation approaches.
- Consumption effects remain negative and stunting effects remain positive across most specifications.

Economic message: The main result is not a fragile specification. The *mita* is associated with persistently lower income and worse human development.

Dependent Variable						
Log Equiv. Household Consumption (2001)			Stunted Growth, Children 6-9 (2005)			
<100 km of Bound. (1)	<75 km of Bound. (2)	<50 km of Bound. (3)	<100 km of Bound. (4)	<75 km of Bound. (5)	<50 km of Bound. (6)	Border District (7)
Panel A. Cubic Polynomial in Latitude and Longitude						
-0.284 (0.198)	-0.216 (0.207)	-0.331 (0.219)	0.070 (0.043)	0.084* (0.046)	0.087* (0.048)	0.114** (0.049)
0.060	0.060	0.069	0.051	0.020	0.017	0.050
Panel B. Cubic Polynomial in Distance to Potosí						
-0.337*** (0.087)	-0.307*** (0.101)	-0.329*** (0.096)	0.080*** (0.021)	0.078*** (0.022)	0.078*** (0.024)	0.063* (0.032)
0.046	0.036	0.047	0.049	0.017	0.013	0.047
Panel C. Cubic Polynomial in Distance to Mita Boundary						
-0.277*** (0.078)	-0.230*** (0.089)	-0.224** (0.092)	0.073*** (0.023)	0.061*** (0.022)	0.064*** (0.023)	0.055* (0.030)
0.044	0.042	0.040	0.040	0.015	0.013	0.043
yes	yes	yes	yes	yes	yes	yes
yes	yes	yes	yes	yes	yes	yes
71	60	52	289	239	185	63
1478	1161	1013	158,848	115,761	100,446	37,421

n is the household in columns 1-3 and the individual in columns 4-7. Robust standard errors, adjusted for clustering by district, are in parentheses. The dependent variable is log imputation (ENAH0 (2003)) in columns 1-3, and a dummy equal to 1 if the child has stunted growth and equal to 0 otherwise in columns 4-7 (Ministerio de Educación (2005a)). Mita is a household's district contributed to the mita and equal to 0 otherwise (Saigues (1984), Amay y Junier (1947, pp. 249, 284)). Panel A includes a cubic polynomial in the latitude and y's district capital, panel B includes a cubic polynomial in Euclidean distance from the observation's district capital to Potosí, and panel C includes a cubic polynomial in Euclidean it on the mita boundary. All regressions include controls for elevation and slope, as well as boundary segment fixed effects (F.E.s). Columns 1-3 include demographic controls for ten, and adults in the household. In columns 1 and 4, the sample includes observations whose district capitals are located within 100 km of the mita boundary, and this threshold in the succeeding columns. Column 1 includes only observations whose districts border the mita boundary, 78% of the observations are in mita districts in column 1, 71% in column 4, 71% in column 5, 68% in column 6, and 58% in column 7. Coefficients that are significantly different from zero are denoted by the following system: *10%, **5%, and

Table II: Living standards

TABLE III—Continued

		Dependent Variable					
		Log Equiv. Household Consumption (2001)			Stunted Growth, Children 6-9 (2005)		
Sample Within:	<100 km of Bound. (1)	<75 km of Bound. (2)	<50 km of Bound. (3)	<100 km of Bound. (4)	<75 km of Bound. (5)	<50 km of Bound. (6)	Border District (7)
Alternative Functional Forms for RD Polynomial: Baseline III							
Linear polynomial in distance to <i>mita</i> boundary							
<i>Mita</i>	-0.299*** (0.082)	-0.227** (0.089)	-0.223** (0.091)	0.072*** (0.024)	0.060*** (0.022)	0.058** (0.023)	0.056* (0.032)
Quadratic polynomial in distance to <i>mita</i> boundary							
<i>Mita</i>	-0.277*** (0.078)	-0.227** (0.089)	-0.224** (0.092)	0.072*** (0.023)	0.060*** (0.022)	0.061*** (0.023)	0.056* (0.030)
Quartic polynomial in distance to <i>mita</i> boundary							
<i>Mita</i>	-0.251*** (0.078)	-0.229** (0.089)	-0.246*** (0.088)	0.073*** (0.023)	0.064*** (0.022)	0.063*** (0.023)	0.055* (0.030)
Interacted linear polynomial in distance to <i>mita</i> boundary							
<i>Mita</i>	-0.301* (0.174)	-0.277 (0.190)	-0.385* (0.210)	0.082 (0.054)	0.087 (0.055)	0.095 (0.065)	0.132** (0.053)
Interacted quadratic polynomial in distance to <i>mita</i> boundary							
<i>Mita</i>	-0.351 (0.260)	-0.505 (0.319)	-0.295 (0.366)	0.140* (0.082)	0.132 (0.084)	0.136 (0.086)	0.121* (0.064)
Ordinary Least Squares							
<i>Mita</i>	-0.294*** (0.083)	-0.288*** (0.089)	-0.227** (0.090)	0.057** (0.025)	0.048* (0.024)	0.049* (0.026)	0.055* (0.031)
Geo. controls	yes	yes	yes	yes	yes	yes	yes
Boundary F.E.s	yes	yes	yes	yes	yes	yes	yes
Clusters	71	60	52	289	239	185	63
Observations	1478	1161	1013	158,848	115,761	100,446	37,421

^a Robust standard errors, adjusted for clustering by district, are in parentheses. All regressions include geographic controls and boundary segment fixed effects (F.E.s).

Table III: specification tests, continued

TABLE III
SPECIFICATION TESTS^a

Dependent Variable							
	Log Equiv. Household Consumption (2001)			Stunted Growth, Children 6-9 (2005)			
Sample Within:	<100 km of Bound. (1)	<75 km of Bound. (2)	<50 km of Bound. (3)	<100 km of Bound. (4)	<75 km of Bound. (5)	<50 km of Bound. (6)	Border District (7)
Alternative Functional Forms for RD Polynomial: Baseline I							
Linear polynomial in latitude and longitude							
<i>Mita</i>	-0.294*** (0.092)	-0.199 (0.126)	-0.143 (0.128)	0.064*** (0.021)	0.054** (0.022)	0.062** (0.026)	0.068** (0.031)
Quadratic polynomial in latitude and longitude							
<i>Mita</i>	-0.151 (0.189)	-0.247 (0.209)	-0.361 (0.216)	0.073* (0.040)	0.091** (0.043)	0.106** (0.047)	0.087** (0.041)
Quartic polynomial in latitude and longitude							
<i>Mita</i>	-0.392* (0.225)	-0.324 (0.231)	-0.342 (0.260)	0.073 (0.056)	0.072 (0.050)	0.057 (0.048)	0.104** (0.042)
Alternative Functional Forms for RD Polynomial: Baseline II							
Linear polynomial in distance to Potosi							
<i>Mita</i>	-0.297*** (0.079)	-0.273*** (0.093)	-0.220** (0.092)	0.050** (0.022)	0.048** (0.022)	0.049** (0.024)	0.071** (0.031)
Quadratic polynomial in distance to Potosi							
<i>Mita</i>	-0.345*** (0.086)	-0.262*** (0.095)	-0.309*** (0.100)	0.072*** (0.023)	0.064*** (0.022)	0.072*** (0.023)	0.060* (0.032)
Quartic polynomial in distance to Potosi							
<i>Mita</i>	-0.331*** (0.086)	-0.310*** (0.100)	-0.330*** (0.097)	0.078*** (0.021)	0.075*** (0.020)	0.071*** (0.021)	0.053* (0.031)
Interacted linear polynomial in distance to Potosi							
<i>Mita</i>	-0.307*** (0.092)	-0.280*** (0.094)	-0.227** (0.095)	0.051** (0.022)	0.048** (0.021)	0.043* (0.022)	0.076*** (0.029)
Interacted quadratic polynomial in distance to Potosi							
<i>Mita</i>	-0.264*** (0.087)	-0.177* (0.096)	-0.285** (0.111)	0.033 (0.024)	0.027 (0.023)	0.039* (0.023)	0.036 (0.024)

(Continues)

Table III: specification tests, first page

TABLE IV
ADDITIONAL SPECIFICATION TESTS^a

Log Equivalent Household Consumption (2001)							Stunted Growth, Children 6-9 (2005)					
	Baseline (1)	Control for Ethnicity (2)	Includes Cusco (3)	Excludes Districts With Inca Estates (4)	Excludes Portions of Boundary Formed by Rivers (5)	Flexible Estimation of Consump. Equivalence (6)	Migration (7)	Baseline (8)	Includes Cusco (9)	Excludes Districts With Inca Estates (10)	Excludes Portions of Boundary Formed by Rivers (11)	Migration (12)
Panel A: Cubic Polynomial in Latitude and Longitude												
u	-0.331 (0.219)	-0.202 (0.157)	-0.465** (0.207)	-0.281 (0.265)	-0.322 (0.215)	-0.326 (0.230)	-0.223 (0.198)	0.087* (0.048)	0.147*** (0.048)	0.093* (0.048)	0.090* (0.048)	0.069 (0.049)
	0.069	0.154	0.104	0.065	0.070	0.292	0.067	0.017	0.046	0.019	0.018	0.016
Panel B: Cubic Polynomial in Distance to Potosi												
u	-0.329*** (0.096)	-0.282*** (0.073)	-0.450*** (0.096)	-0.254*** (0.101)	-0.376*** (0.114)	-0.328*** (0.099)	-0.263*** (0.095)	0.078*** (0.024)	0.146*** (0.030)	0.077*** (0.026)	0.081*** (0.024)	0.060** (0.025)
	0.047	0.140	0.087	0.036	0.049	0.275	0.042	0.013	0.039	0.014	0.013	0.012
Panel C: Cubic Polynomial in Distance to Mita Boundary												
u	-0.224** (0.092)	-0.195*** (0.070)	-0.333*** (0.087)	-0.255** (0.110)	-0.217** (0.098)	-0.224** (0.095)	-0.161* (0.088)	0.064*** (0.023)	0.132*** (0.027)	0.066*** (0.025)	0.065*** (0.023)	0.046* (0.024)
	0.040	0.135	0.088	0.047	0.039	0.270	0.037	0.013	0.042	0.014	0.013	0.012
o. controls	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
und. F.E.s	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
clusters	52	52	57	47	51	52	52	185	195	180	183	185
users	1013	1013	1173	930	992	1013	997	100,446	127,259	96,440	99,940	98,922

^a Robust standard errors, adjusted for clustering by district, are in parentheses. All regressions include soil type indicators and boundary segment fixed effects (F.E.s). Columns 1-5 and 7 are demographic controls for the number of infants, children, and adults in the household. Column (8) includes controls for the log of household size and the ratio of children to household users, using the log of household consumption as the dependent variable. The samples include observations whose district capitals are less than 40 km from the mita boundary. Coefficients

Table IV: additional specification tests

7.4 Figure 2: Spatial RD plots of outcomes

Read:

- Figure 2 plots outcomes against longitude and latitude, with predicted values from spatial RD specifications.
- Panels show consumption, stunting, hacienda measures, education, road density, and market participation.
- The spatial plots visually summarize how outcomes change around the boundary.

Economic message: The figure provides a geographic check that the estimated discontinuities are not merely abstract regression coefficients. The mita boundary lines up with visible differences in development channels.

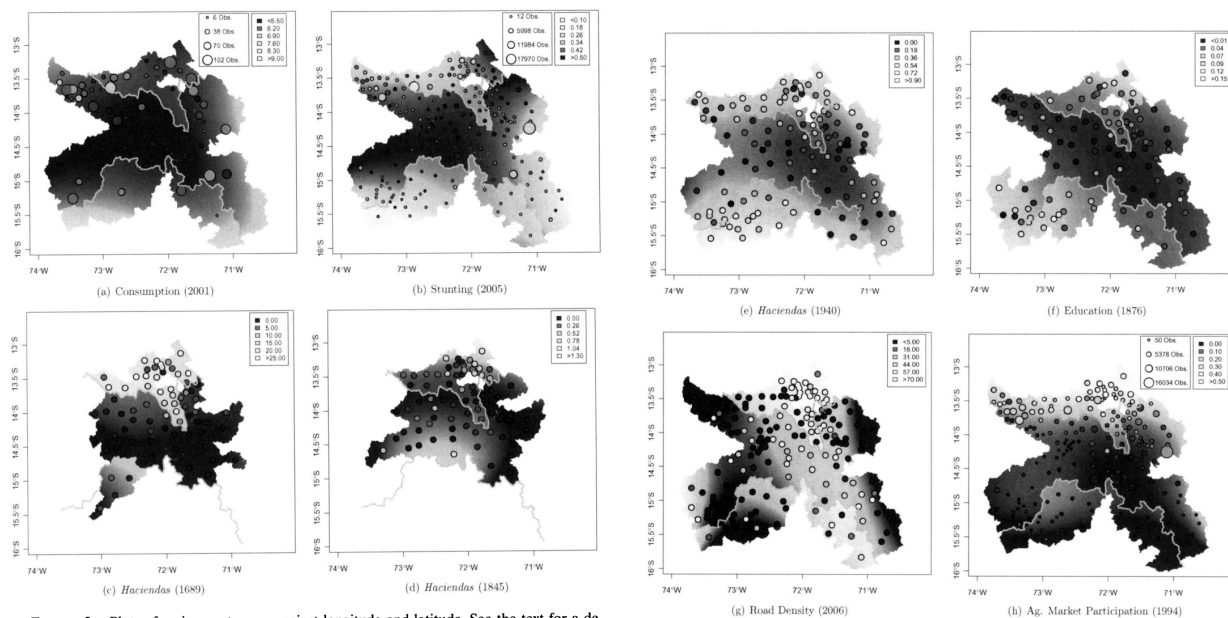


FIGURE 2.—Plots of various outcomes against longitude and latitude. See the text for a detailed description.

Figure 2, panels a-d

FIGURE 2.—Continued.

Figure 2, panels e-h

7.5 Table V and Table VI: Pre-treatment data and land tenure

Read:

- Table V uses 1572 tribute and population data and finds no major pre-treatment differences across the mita boundary.
- Table VI shows that mita districts had substantially fewer haciendas historically.
- The negative hacienda effects appear in 1689 and persist in later hacienda population measures.
- The land Gini in 1994 is higher in mita districts, consistent with altered land tenure and property-rights evolution.

Economic message: The paper's mechanism starts with colonial land tenure: the mita discouraged the formation of large estates inside the treated region, which changed political incentives and public-goods provision.

	Dependent Variable							
	Share of Tribute Revenues					Percent		
	Log Mean Tribute (1)	Spanish Nobility (2)	Spanish Priests (3)	Spanish Justices (4)	Indig. Mayors (5)	Men (6)	Boys (7)	Females (8)
Panel A. Cubic Polynomial in Latitude and Longitude								
<i>Mita</i>	0.020 (0.031)	-0.010 (0.030)	0.004 (0.019)	0.004 (0.010)	0.003 (0.005)	-0.006 (0.009)	0.011 (0.012)	-0.009 (0.016)
<i>R</i> ²	0.762	0.109	0.090	0.228	0.266	0.596	0.377	0.599
Panel B. Cubic Polynomial in Distance to Potosí								
<i>Mita</i>	0.019 (0.029)	-0.013 (0.025)	0.008 (0.015)	0.006 (0.009)	-0.001 (0.004)	-0.012 (0.008)	0.005 (0.010)	-0.011 (0.012)
<i>R</i> ²	0.597	0.058	0.073	0.151	0.132	0.315	0.139	0.401
Panel C. Cubic Polynomial in Distance to <i>Mita</i> Boundary								
<i>Mita</i>	0.040 (0.030)	-0.009 (0.018)	0.005 (0.012)	0.003 (0.006)	-0.001 (0.004)	-0.011 (0.007)	0.001 (0.008)	-0.008 (0.010)
<i>R</i> ²	0.406	0.062	0.096	0.118	0.162	0.267	0.190	0.361
Geo. controls	yes	yes	yes	yes	yes	yes	yes	yes
Boundary F.E.s	yes	yes	yes	yes	yes	yes	yes	yes
Mean dep. var.	1.591	0.625	0.203	0.127	0.044	0.193	0.204	0.544
Observations	65	65	65	65	65	65	65	65

^aThe dependent variable in column 1 is the log of the district's mean 1572 tribute rate (Miranda (1583)). In

Table V: 1572 tribute and population

	Dependent Variable				
	<i>Haciendas</i> per District in 1689 (1)	<i>Haciendas</i> per 1000 District Residents in 1689 (2)	Percent of Rural Tributary Population in <i>Haciendas</i> in ca. 1845 (3)	Percent of Rural Population in <i>Haciendas</i> in 1940 (4)	Land Gini in 1994 (5)
Panel A. Cubic Polynomial in Latitude and Longitude					
<i>Mita</i>	-12.683*** (3.221)	-6.453** (2.490)	-0.127* (0.067)	-0.066 (0.086)	0.078 (0.053)
<i>R</i> ²	0.538	0.582	0.410	0.421	0.245
Panel B. Cubic Polynomial in Distance to Potosí					
<i>Mita</i>	-10.316*** (2.057)	-7.570*** (1.478)	-0.204** (0.082)	-0.143*** (0.051)	0.107*** (0.036)
<i>R</i> ²	0.494	0.514	0.308	0.346	0.194
Panel C. Cubic Polynomial in Distance to <i>Mita</i> Boundary					
<i>Mita</i>	-11.336*** (2.074)	-8.516*** (1.665)	-0.212*** (0.060)	-0.120*** (0.045)	0.124*** (0.033)
<i>R</i> ²	0.494	0.497	0.316	0.336	0.226
Geo. controls	yes	yes	yes	yes	yes
Boundary F.E.s	yes	yes	yes	yes	yes
Mean dep. var.	6.500	5.336	0.135	0.263	0.783
Observations	74	74	81	119	181

^aThe unit of observation is the district. Robust standard errors are in parentheses. The dependent variable in column 1 is *haciendas* per district in 1689 and in column 2 is *haciendas* per 1000 district residents in 1689 (Villanueva Uribeaga (1982)). In column 3 it is the percentage of the district's tributary population residing in *haciendas* ca. 1845 (Peralta Ruiz (1991)), in column 4 it is the percentage of the district's rural population residing in *haciendas* in 1940 (Dirección de Estadística del Perú (1944)), and in column 5 it is the district land gini (INEI (1994)). Panel A includes a cubic polynomial in the latitude and longitude of the observation's district capital, panel B includes a cubic polynomial in Euclidean distance from the observation's district capital to Potosí, and panel C includes a cubic polynomial in Euclidean distance to the nearest point on the *mita* boundary. All regressions include geographic controls and boundary segment fixed effects. The samples include districts whose capitals are less than 50 km from the *mita* boundary. Column 3 is weighted by the square root of the district's rural tributary population and column 4 is weighted by the square root of the district's rural population. 58% of the observations are in *mita* districts in column 1 and 2. 50% in

Table VI: land tenure and labor systems

7.6 Table VII and Table VIII: Education and roads

Read:

- Table VII shows that the *mita* lowered historical literacy and schooling in some specifications, particularly in 1876 and 1940.
- Table VIII shows large negative effects on regional road networks and paved/gravel regional roads.
- The *mita* has weaker effects on local roads, but strong effects on broader integration into regional transport networks.

Economic message: These tables connect land tenure to public goods. Areas with weaker property rights and weaker elite incentives appear to have received less education and transport infrastructure.

TABLE VII
EDUCATION^a

	Dependent Variable		
	Literacy 1876 (1)	Mean Years of Schooling 1940 (2)	Mean Years of Schooling 2001 (3)
Panel A. Cubic Polynomial in Latitude and Longitude			
<i>Mita</i>	-0.015 (0.012)	-0.265 (0.177)	-1.479* (0.872)
R^2	0.401	0.280	0.020
Panel B. Cubic Polynomial in Distance to Potosí			
<i>Mita</i>	-0.020*** (0.007)	-0.181** (0.078)	-0.341 (0.451)
R^2	0.345	0.187	0.007
Panel C. Cubic Polynomial in Distance to <i>Mita</i> Boundary			
<i>Mita</i>	-0.022*** (0.006)	-0.209*** (0.076)	-0.111 (0.429)
R^2	0.301	0.234	0.004
Geo. controls	yes	yes	yes
Boundary F.E.s	yes	yes	yes
Mean dep. var.	0.036	0.470	4.457
Clusters	95	118	52
Observations	95	118	4038

^aThe unit of observation is the district in columns 1 and 2 and the individual in column 3. Robust standard errors, adjusted for clustering by district, are in parentheses. The dependent variable is mean literacy in 1876 in column 1 (Dirección de Estadística del Perú (1878)), mean years of schooling in 1940 in column 2 (Dirección de Estadística del Perú (1944)), and individual years of schooling in 2001 in column 3 (ENAH0 (2001)). Panel A includes a cubic polynomial in the latitude and longitude of the observation's district capital, panel B includes a cubic polynomial in Euclidean distance from the observation's district capital to Potosí, and panel C includes a cubic polynomial in Euclidean distance to the nearest point on the *mita* boundary. All regressions include geographic controls and boundary segment fixed effects. The samples include districts whose capitals are less than 50 km from the *mita* boundary. Columns 1 and 2 are weighted by the square root of the district's population. 64% of the observations are in *mita*

Table VII: education

TABLE VIII
ROADS^a

	Dependent Variable		
	Density of Local Road Networks (1)	Density of Regional Road Networks (2)	Density of Paved/Gravel Regional Roads (3)
Panel A. Cubic Polynomial in Latitude and Longitude			
<i>Mita</i>	0.464 (18.575)	-29.276* (16.038)	-22.426* (12.178)
R^2	0.232	0.293	0.271
Panel B. Cubic Polynomial in Distance to Potosí			
<i>Mita</i>	-1.522 (12.101)	-32.644*** (8.988)	-30.698*** (8.155)
R^2	0.217	0.271	0.256
Panel C. Cubic Polynomial in Distance to <i>Mita</i> Boundary			
<i>Mita</i>	0.535 (12.227)	-35.831*** (9.386)	-32.458*** (8.638)
R^2	0.213	0.226	0.208
Geo. controls	yes	yes	yes
Boundary F.E.s	yes	yes	yes
Mean dep. var.	85.34	33.55	22.51
Observations	185	185	185

^aThe unit of observation is the district. Robust standard errors are in parentheses. The road densities are defined as total length in meters of the respective road type in each district divided by the district's surface area, in kilometers squared. They are calculated using a GIS map of Peru's road networks (Ministro de Transporte (2006)). Panel A includes a cubic polynomial in the latitude and longitude of the observation's district capital, panel B includes a cu-

Table VIII: roads

7.7 Table IX and Figure 3: Consumption channels and market participation

Read:

- Table IX shows that mita districts are more likely to rely on agriculture and less likely to sell agricultural output in markets.
- The coefficient on market sales is negative and significant across the preferred specifications.
- Figure 3 shows predicted values for market participation under alternative spatial specifications.

Economic message: The modern channel is reduced market integration. The mita's institutional legacy appears in a higher prevalence of subsistence farming and lower participation in agricultural markets.

TABLE IX
CONSUMPTION CHANNELS^a

	Dependent Variable		
	Percent of District Labor Force in Agriculture—1993 (1)	Agricultural Household Sells Part of Produce in Markets—1994 (2)	Household Member Employed Outside the Agricultural Unit—1994 (3)
Panel A. Cubic Polynomial in Latitude and Longitude			
<i>Mita</i>	0.211 (0.140)	−0.074** (0.036)	−0.013 (0.032)
R^2	0.177	0.176	0.010
Panel B. Cubic Polynomial in Distance to Potosí			
<i>Mita</i>	0.101 (0.061)	−0.208*** (0.030)	−0.033 (0.020)
R^2	0.112	0.144	0.008
Panel C. Cubic Polynomial in Distance to <i>Mita</i> Boundary			
<i>Mita</i>	0.092* (0.054)	−0.225*** (0.032)	−0.038** (0.018)
R^2	0.213	0.136	0.006
Geo. controls	yes	yes	yes
Boundary F.E.s	yes	yes	yes
Mean dep. var.	0.697	0.173	0.245
Clusters	179	178	182
Observations	179	160,990	183,596

^aRobust standard errors, adjusted for clustering by district in columns 2 and 3, are in parentheses. The dependent variable in column 1 is the percentage of the district's labor force engaged in agriculture as a primary occupation (INEI (1993)); in column 2 it is an indicator equal to 1 if the agricultural unit sells at least part of its produce in markets, and in column 3 it is an indicator equal to 1 if at least one member of the household pursues secondary employment outside the agricultural unit (FAO/IFAD 2004).

Table IX: consumption channels

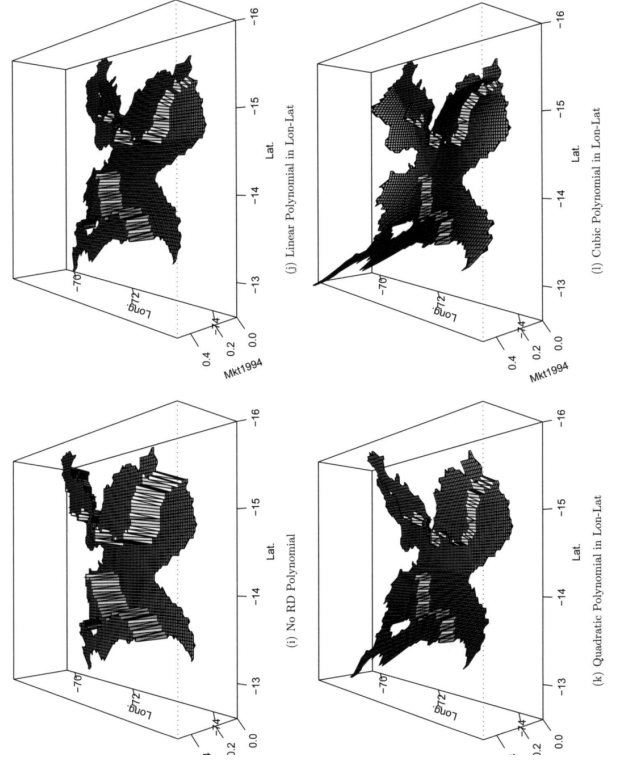


Figure 3: market participation surfaces

8 Short summary

- The paper studies the long-run effects of Peru's colonial mining mita.
- The mita was a forced labor draft used from 1573 to 1812.
- Treatment is assigned by a historical geographic boundary, enabling a regression discontinuity design.
- The main outcomes show lower modern household consumption and higher child stunting in mita districts.
- Balance tests using geography, ethnicity, and 1572 tribute data support the validity of the RD design.
- The paper identifies land tenure and public goods as channels of persistence.
- Mita districts historically had fewer haciendas and lower educational attainment.
- Today, mita districts have weaker regional road networks and less market-oriented agriculture.
- The paper contributes micro-level causal evidence to the literature on colonial institutions and long-run development.

9 Conclusion

9.1 Core conclusion

Dell shows that a coercive colonial labor institution had large and persistent effects on modern economic outcomes. The mita lowered household consumption and worsened child health nearly two centuries after its abolition.

9.2 Economic intuition

The institution affected more than labor allocation during the colonial period. It changed the organization of landholding and the incentives for public-goods provision. Areas outside the mita developed haciendas with secure property rights and politically connected landowners, while mita districts retained communal land tenure and later faced weaker property-rights enforcement. These institutional paths shaped roads, education, market participation, and poverty.

9.3 Takeaway

The paper’s lasting insight is that historical institutions persist through concrete local mechanisms. The mining mita mattered not simply because it extracted labor, but because it reshaped the political economy of land, infrastructure, and market access. The study is therefore a benchmark example of how to combine historical research with modern causal inference.